

Uncovering Hidden Ion Channels in the Proteome using AI and Molecular Simulations

1. Introduction

- **Ion channels** are integral membrane proteins that enable **ion transport** across membranes.
- Many **transmembrane (TMEM) proteins** remain uncharacterised and may function as ion channels.¹
- **Dysfunction** is linked to diseases including **cancer**, **Parkinson's** and **Alzheimer's**.²
- **AIM:** To assess whether **TMEM43** forms an ion-conducting pore using **AI-based structural prediction** and **computational pore analysis**.

2. Methodology

As the oligomeric state of TMEM43 is unknown, **monomeric, dimeric and trimeric** structures were evaluated.

TMEM43 sequence retrieved from UniProt.

Structures were predicted using **AlphaFold2** via the ColabFold server.

Structures were oriented within a membrane using the **PPM** server prior to pore analysis through **CHAP**.³

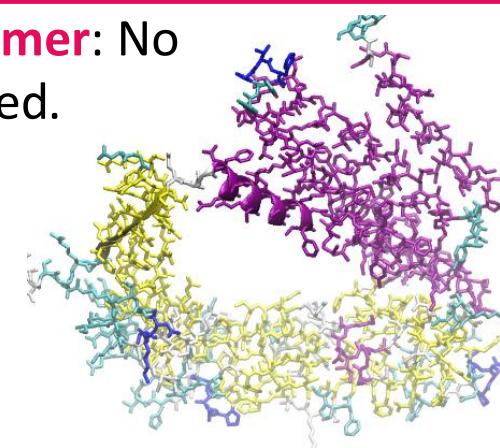
The oligomers were visualised in Visual Molecular Dynamics (**VMD**).⁴

Although predicting larger oligomers can take **3+ hours**, this computational approach is still less time-consuming than experimental characterisation.

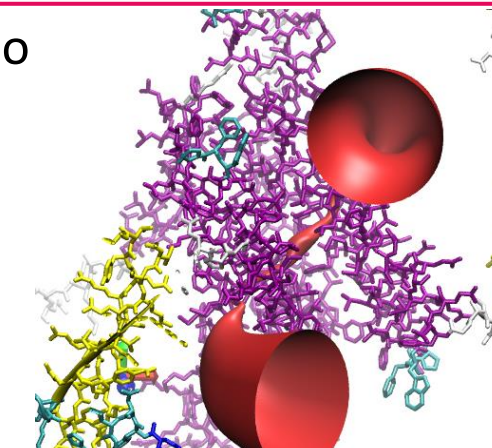
3. Results

a. Using VMD to identify any pores and visualize permeation pathways:

TMEM43 **Monomer**: No pathway detected.

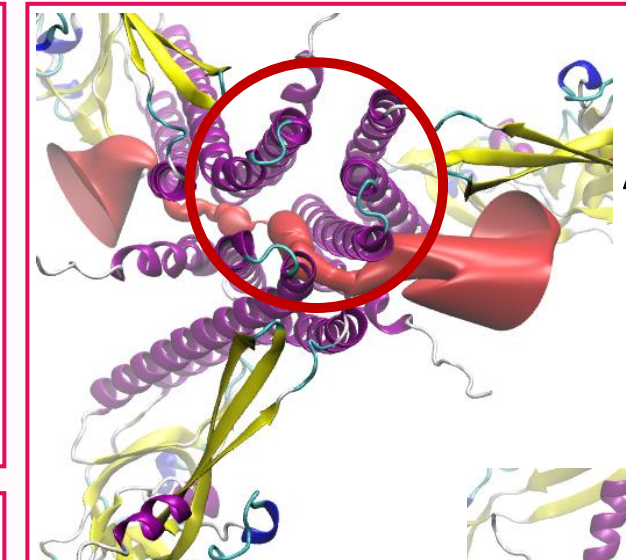


TMEM43 **Dimer**: No pathway detected

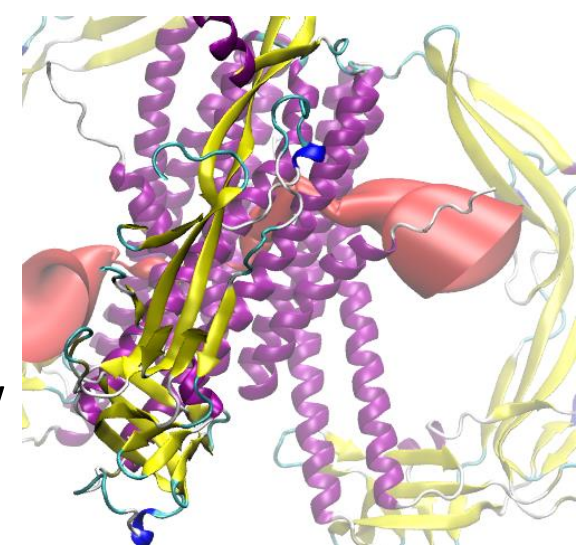


TMEM43 **Trimer**:

A more promising structure with a pore-like channel entrance.



However, the **permeation pathway** (colored red) is too narrow for an ion to pass through.



b. Pore radius and hydrophobicity profile of the trimer

The minimum pore radius is **~0.16 Å**, where the lining residues are **moderately hydrophobic**.

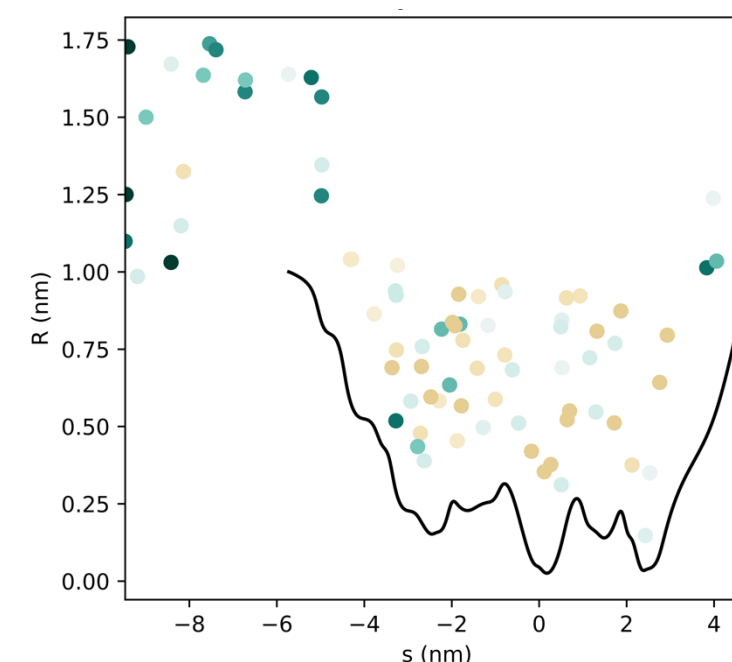


Fig. 1 Radius profile of TMEM43 trimer

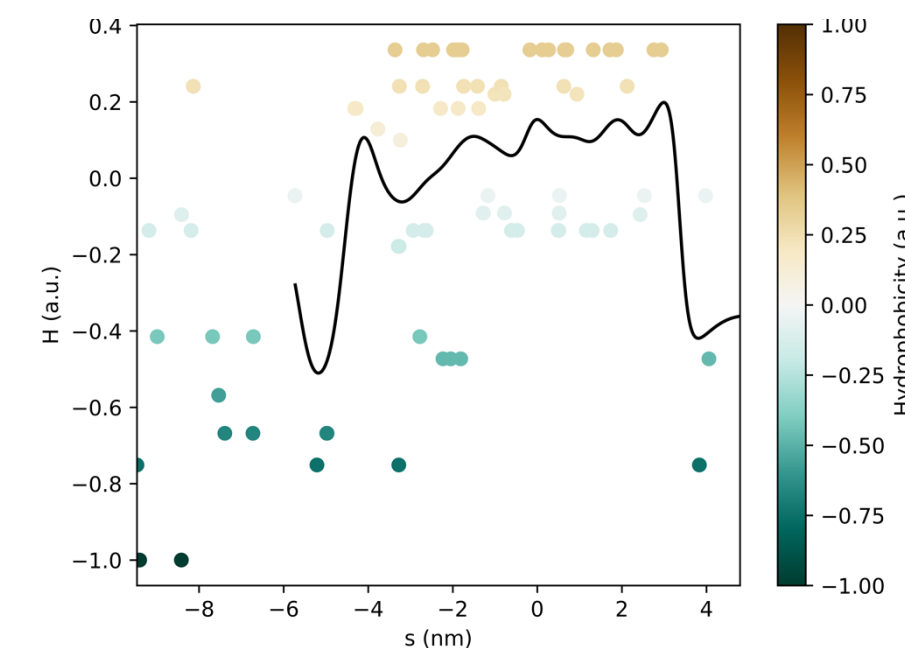


Fig. 2 Hydrophobicity profile of TMEM43 trimer

CHAP analysis was used to assess potential **hydrophobic gating**, which can regulate ion permeation and selectivity in ion channels.

4. Conclusions

- Structures predicted for TMEM43 monomer, dimer and trimer.
- The minimum pore radius (**~0.16 Å**) is far smaller than the radius expected for hydrated ions (**~3 Å**) and the radius of water molecules (**~1.4 Å**).
- Therefore, even though there seems to be a pore, **TMEM43 is unlikely to be conductive** in this conformation.

5. Future work

- Predict a structure for TMEM43 **tetramer**.
- Select a new uncharacterized TMEM (i.e. TMEM120A) if negative results persist.
- Trimer equilibration using **MD simulations** in an explicit lipid membrane.

References:

1. B. Hille, *Biophys. J.*, 1978, **22**, 283–294.
2. H. Kang and C. J. Lee, *Springer Nature*, 2024, preprint, DOI: 10.1038/s12276-024-01206-1.
3. G. Klesse, S. Rao, M. S. P. Sansom and S. J. Tucker, *J. Mol. Biol.*, 2019, **431**, 3353–3365.
4. W. Humphrey, A. Dalke and K. Schulten, *J. Mol. Graph.*, 1996, **14**, 33–38.